GUIDE: Deconstructing Genetic Architecture using Independent Latent-Factor Decomposition

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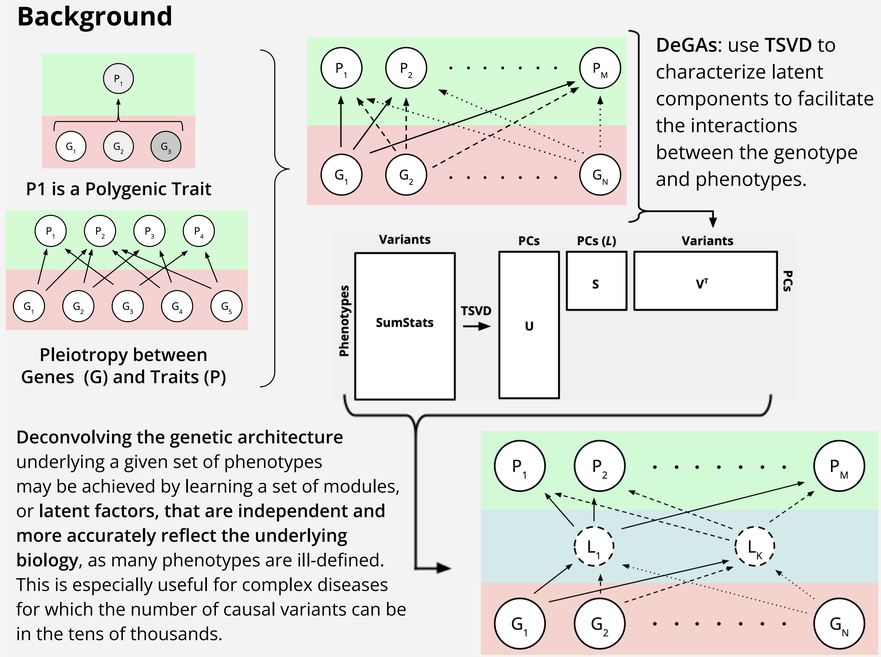
Background



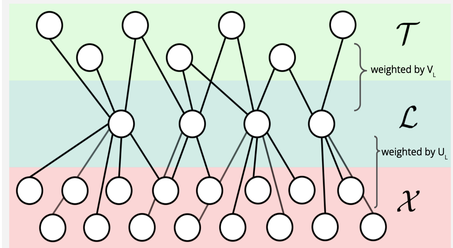
Deconvolving the genetic architecture underlying a given set of phenotypes may be achieved by learning a set of modules, or **latent factors, that are independent and more accurately reflect the underlying biology**, as many phenotypes are ill-defined. This is especially useful for complex diseases for which the number of causal variants can be in the tens of thousands.

ICA: Compute an Informed Basis

Let B be the matrix of effect size (beta) summary statistics. The respective SVD and TSVD decompositions can be expressed as:

$$B = USV^T \qquad B \approx U_L S_L V_L^T$$

which yields the following network structure:



We examine the weights and **introduce a rotation and scaling matrix, A, to account for the proper scaling and orientation of the latent components** such that:

$$A \in \mathrm{GL}(L,\mathbb{R}) \implies U_L V_L^T = U_L A A^{-1} V_L^T$$

By whitening and treating scaling separately, we can instead look for

$$A \in \mathrm{O}(L,\mathbb{R})$$

we redefine the weight matrices (assuming a proper basis has been found):

$$W_{\mathcal{XL}} = U_L A$$
$$W_{\mathcal{LT}} = A^T V_L^T$$

Thus, we can rewrite the decomposition as the following after transformation under the **latent basis**:

$$\hat{B} = \underbrace{UA}_{\substack{\text{normalized}\\\text{weights in}\\\text{new basis}}} \overbrace{A^T S A}^{\substack{\text{scaling relative}\\\text{to new basis}}} \underbrace{A^T V^T}_{\substack{\text{normalized}\\\text{weights in}\\\text{new basis}}}$$

A acts as a permutation matrix within the new basis. Thus, the network can be represented as:

$$\hat{B} = \overbrace{UA}^{BW^T_{\mathcal{LT}}} \underbrace{A^T S A A^T V^T}_{W^T_{\mathcal{XL}} B}$$

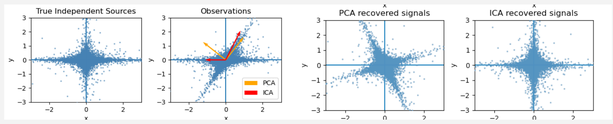
Given a set of SNPs X and traits T, we compute a **basis** of L, latent traits, that **render the distributions of W_XL and W_LT maximally non-Gaussian**.

Minimize Entropy (or similar):

Theorem 1 (Entropy increases at every step). *Let $X_1, X_2, \ldots$ be independent and identically distributed square-integrable random variables. Then:*

$$\mathrm{Ent}\left(\frac{X_1 + \ldots + X_n}{\sqrt{n}}\right) \le \mathrm{Ent}\left(\frac{X_1 + \ldots + X_n + 1}{\sqrt{n+1}}\right)$$

Following the infinitesimal model, weights for biologically interpretable latent traits will not be perfectly Gaussian. When this holds, any rotation that mixes up the latent components yields more Gaussian weights. **So ICA finds the basis representing the rotation in the latent space that maximizes non-Gaussianity: a representation maximizing sparsity.**

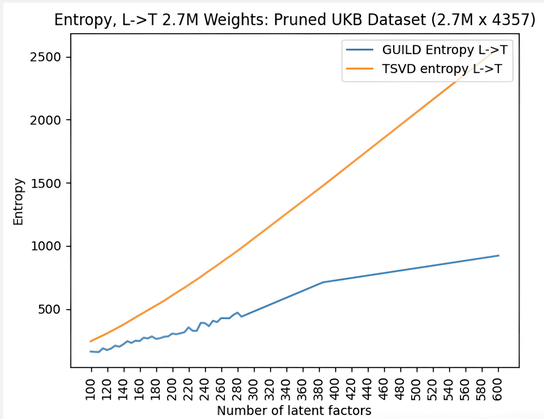


$$A = \underset{\Lambda \in \mathrm{GL}(L,\mathbb{R})}{\mathrm{argmin}} \left(\mathrm{Ent}(U\Lambda^{-1}) + \mathrm{Ent}(\Lambda V^T)\right) \qquad A = \underset{\Lambda \in \mathrm{GL}(L,\mathbb{R})}{\mathrm{argmax}} \left(\mathrm{Kurt}(U\Lambda^{-1}) + \mathrm{Kurt}(\Lambda V^T)\right)$$

Our Method: Genetic Un-mixing by Independent DEcomposition

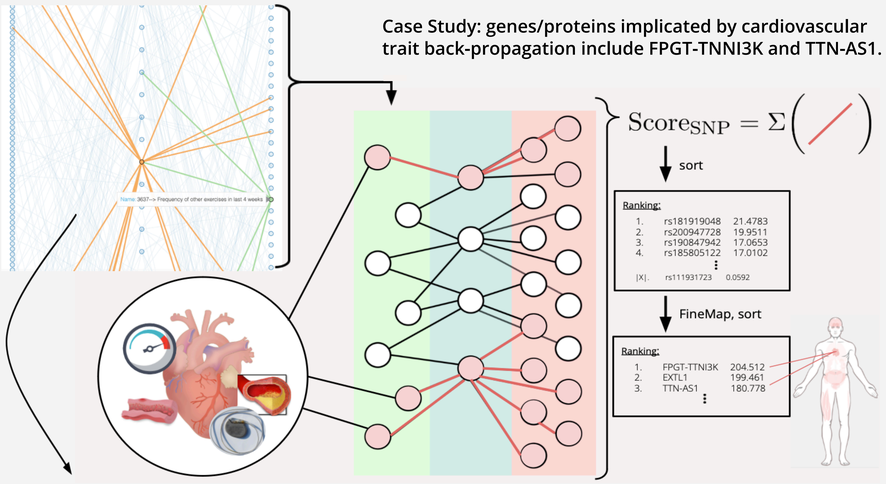
Algorithm 1 GUIDE: TSVD + ICA

Data: B, a SNPs × Traits matrix of effect sizes (β's)
Result: Unmixing matrix A, $W_{\mathcal{XL}}$,$W_{\mathcal{LT}}$, L
$B \leftarrow C_{\text{SNPs}} B C_{\text{Traits}}$ ▷ Apply mean-centering matrices to B
$U, S, V^T \leftarrow \text{SVD}(B, L)$ ▷ Compute the TSVD of B
$\mathcal{G} \leftarrow (U^T | V^T)$
$\mathcal{G}_u = A\mathcal{G}$ ▷ Find unmixing matrix A using ICA
return $W_{\mathcal{XL}} \leftarrow UA, \quad W_{\mathcal{LT}} \leftarrow A^T V^T$



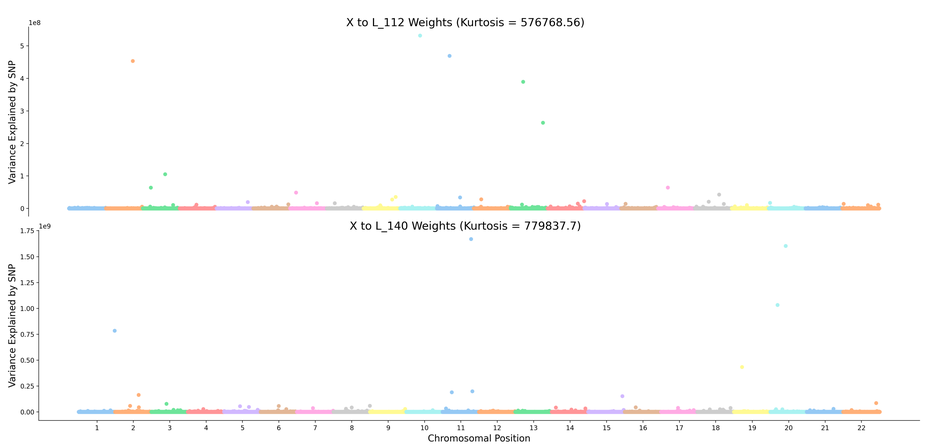
GUIDE on pruned UKBB vs. TSVD alone exhibits higher info gained

Potential Application: Exploratory Analysis

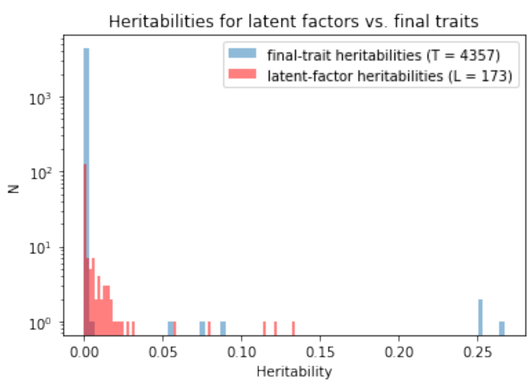


	Latent factor number	X –> L weight	SNP number	SNP ID
0	1	0.543613	175	4:100239319
1	1	0.150106	780	11:48266736
2	1	0.115433	954	14:50655357
3	1	0.149013	956	14:51207741
4	2	-0.137323	39	1:113161350
...	...	...	...	...

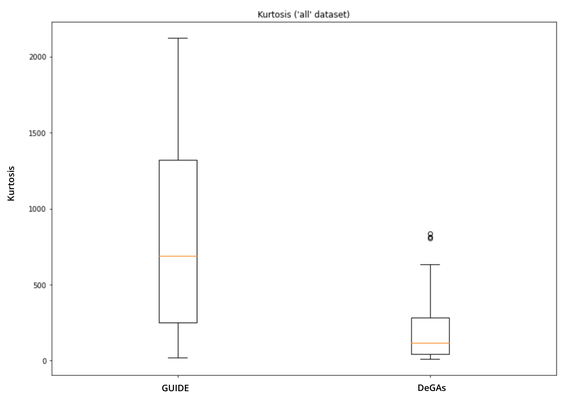
	Latent factor number	L –> T weight	trait number	trait ID
0	1	0.934885	213	20414--> Frequency of drinking alcohol
1	2	-0.917244	182	5103_irnt--> 3mm weak meridian angle (left)
2	4	-0.963974	318	4283--> Number of rounds of numeric memory tes...
3	6	0.956087	148	3536_irnt--> Age started hormone-replacement t...
4	7	-0.953061	95	6033_irnt--> Maximum heart rate during fitness...
5	8	-0.946179	102	5119_irnt--> 3mm cylindrical power (left)
6	12	0.945938	294	5100_irnt--> 3mm weak meridian angle (right)
7	13	0.983690	113	4270--> Volume level set by participant (left)



Demonstrating the sparsity induced by GUIDE; a few SNPs from the same chromosome account for significant variance.



A comparison of heritabilities for final traits vs. those for the latent factors. Latent factors capture more heritability on average and the same total heritability for all traits.



A comparison between our method and DeGAs on their curated 'All' dataset (# SNPs = 235,907, # Traits = 2,138), demonstrating higher kurtosis, or more information gained.

Conclusions

1. **GUIDE** uses **TSVD** and **ICA** to estimate a set of **statistically independent latent factors** that best express the information encoded in the underlying genetic architecture, constrained by genetic data.
2. GUIDE's latent components **provide unbiased and data-driven SNP, gene, and trait groupings** that are **independent of our prior phenotypic definitions** for downstream analysis.